

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1703

COURSE OUTCOME COVERED

***CO4:** Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I I. MD IMTHIAZ / 192521360 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

I. IMTHIAZ

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Answer:

(a) Article Citation, Domain and ML Problem

Full citation (APA style): Deberneh, H. M., & Kim, I. (2021). Prediction of Type 2 Diabetes Based on Machine Learning Algorithm. *International Journal of Environmental Research and Public Health*, 18(6), 3317. DOI: 10.3390/ijerph18063317.

Domain / application area: The article belongs to the healthcare and clinical decision-support domain, specifically preventive medicine and chronic-disease risk prediction using electronic health records (EHRs).

Specific ML problem addressed: The authors address a multi-class classification / forward-prediction problem. Instead of only diagnosing a patient's current status, the model predicts, one year in advance, whether a person will be Normal, Prediabetic, or Diabetic in the following year, using that person's current-year clinical and lifestyle attributes. This is a forecasting task built on supervised, labelled historical records rather than a simple present-time diagnosis.

(b) Summary of Objective and Research Gap (150-200 words)

The authors aim to build a machine-learning model that forecasts the onset of type 2 diabetes (T2D) one year ahead, classifying a patient's future state as normal, prediabetic, or diabetic rather than only diagnosing the present condition. Using six years of electronic health records from a South Korean medical institute, they apply a data-driven feature-selection pipeline (ANOVA test, chi-squared test, and recursive feature elimination) to shortlist twelve clinically relevant predictors, then compare logistic regression, random forest, support vector machine, XGBoost, and three ensemble strategies. The research gap they identify is twofold: most earlier studies only screened for existing diabetes or prediabetes rather than predicting future onset, and very few models attempted to separately predict the prediabetic stage, which is harder to classify because it overlaps statistically with both the normal and diabetic classes. The authors fill this gap by treating the problem as a three-class forward-prediction task and by testing whether ensemble learning and additional years of historical data improve the model's ability to correctly separate the overlapping prediabetes class from the other two.

(c) Article Access Details

This article is published under an open-access Creative Commons Attribution (CC BY) licence by MDPI, and the full text with DOI link is available at <https://doi.org/10.3390/ijerph18063317>. A PDF copy can be downloaded directly from the journal website or from PubMed Central (PMC8004981).

Task 2: Methodology Analysis

- (a)** Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b)** Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Answer:

(a) ML Technique(s) Used and Dataset Description

The core statistical learning model used in the article is Logistic Regression (LR), which the authors treat as the baseline representing the existing clinical statistical-analysis approach. Alongside LR, they train Random Forest (RF), Support Vector Machine (SVM), and XGBoost as comparison models, and combine all of these into three ensemble techniques: a confusion-matrix-based classifier integration model (CIM), a stacking classifier, and soft voting. This mixture allows a direct comparison between a classical statistical learner (Logistic Regression) and several tree-based or margin-based statistical/ML learners on the same clinical prediction task.

Dataset: The data is a six-year (2013-2018) set of electronic health records collected at the Hanaro Medical Foundation in Seoul, South Korea. The raw collection has 535,169 instances from 253,395 subjects, with 1,444 raw features per record, combining numerical laboratory results (such as fasting plasma glucose, BMI, triglycerides) with categorical questionnaire answers (such as smoking, drinking, and family history). After excluding subjects already diagnosed with diabetes, hypertension, or hyperlipidemia and removing records with missing values, the authors applied a three-stage feature-selection pipeline (ANOVA for numeric features, chi-squared test for categorical features, followed by tree-based recursive feature elimination) to reduce the feature set to twelve clinically meaningful predictors: fasting plasma glucose, HbA1c, triglycerides, BMI, gamma-GTP, age, uric acid, sex, smoking, drinking, physical activity, and family history. Because the normal class made up 68.1 percent of the data and diabetes only 4.3 percent, the authors applied majority under-sampling and SMOTE (Synthetic Minority Over-sampling Technique) to correct the class imbalance before training and testing the models with roughly 17,131 training records and 200 held-out test records per class.

(b) Mapping to CO4 Topics and Strength/Limitation

The article maps most directly onto the Statistical Learning Methods strand of CO4, since Logistic Regression, the paper's representative statistical baseline, is directly compared against machine-learning classifiers on the same clinically labelled dataset; the authors' feature-selection and evaluation pipeline (ANOVA, chi-squared, accuracy/precision/recall/F1) mirrors exactly the workflow taught for building and validating a statistical/ML classifier. The article also touches the Decision Tree strand indirectly, since the related-work section reports a competing C5.0 decision-tree model that reached 77.87 percent accuracy on a similar diabetes-prediction task, giving a useful decision-tree benchmark for comparison.

Methodological strength: A clear strength is the rigorous, data-driven feature-selection procedure, which combines a statistical test (ANOVA), a categorical significance test (chi-squared), and a model-based method (recursive feature elimination) rather than relying on features chosen purely by clinical intuition; this reduces the risk of overfitting to irrelevant variables and is validated further by comparing the resulting 12-feature model against the traditional 5-feature clinical predictor set, showing a measurable accuracy improvement.

Methodological limitation: A clear limitation is that the ground-truth diabetes label was defined using fasting plasma glucose (FPG) alone, without HbA1c or an oral glucose tolerance test (OGTT), even though HbA1c was itself used as a predictive feature; this creates a slight inconsistency between the diagnostic criterion and the input features, and the authors themselves acknowledged that the model was developed and validated on a single institution's data, which limits how confidently the results can be generalised to other populations or health systems.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Answer:

(a) Key Results

The authors report accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), and Cohen's kappa (KC) for each model on the held-out test dataset. Since multiple models were benchmarked, the results are summarised in the comparative table below.

Model	Accuracy	Precision	Recall	F1-score	MCC	Kappa
Logistic Regression (LR)	0.71	0.71	0.71	0.71	0.56	0.56
Random Forest (RF)	0.73	0.74	0.73	0.74	0.60	0.60
XGBoost	0.72	0.74	0.72	0.73	0.58	0.58
SVM	0.73	0.74	0.74	0.74	0.60	0.60
Classifier Integration (CIM)	0.73	0.73	0.73	0.73	0.59	0.59
Stacking Classifier	0.72	0.75	0.72	0.73	0.58	0.58
Soft Voting	0.73	0.74	0.73	0.73	0.59	0.59

The confusion matrix for the Random Forest model further shows that most misclassifications occurred around the Prediabetes class, with 58 prediabetic cases wrongly predicted as normal and 16 wrongly predicted as diabeticThe authors also report that cross-validation accuracy rose from about 78 percent (using one year of history) to about 81 percent when four years of historical records were used to train the model, and that the 12-feature data-driven set consistently outperformed the traditional 5-feature clinical predictor set (FPG, HbA1c, BMI, age, sex).

(b) Critical Evaluation

The reported metrics (accuracy of roughly 71-73 percent on the held-out test set) are realistic and clinically believable for a three-class forecasting problem with substantial class overlap; the authors do not inflate their results, and they openly discuss the limiting effect of the overlapping prediabetes class rather than hiding it. However, several potential sources of bias should be noted. First, the dataset comes from a single private medical institute in Seoul, so the model may be biased toward the demographic and lifestyle patterns of that specific South Korean population and may not transfer well to other ethnic groups, dietary habits, or healthcare systems. Second, records with missing values were simply excluded rather than imputed, which could systematically remove subjects with irregular check-up patterns and bias the remaining sample toward more health-conscious individuals. Third, the class-imbalance correction (SMOTE plus under-sampling) was applied before evaluation in a way that is not fully detailed, and if synthetic minority samples leaked into the test set, the reported performance could be mildly optimistic.

To strengthen the findings, additional experiments would help: external validation on an independent, multi-institution or multi-country dataset to test generalisability; inclusion of HbA1c-based or OGTT-based labelling alongside FPG to check label consistency; and an ablation study isolating the individual contribution of each of the twelve selected features to confirm that none of them are redundant. A formal SMOTE-then-split versus split-then-SMOTE comparison would also help verify that the test set was not contaminated by synthetic data.

(c) Real-World Applicability

This forward-prediction approach is well suited for deployment inside hospital electronic health record systems, corporate or insurer-run annual health check-up programmes, and public health screening initiatives, where it could flag patients who are likely to progress from normal to prediabetic or from prediabetic to diabetic within a year, prompting earlier lifestyle counselling or clinical follow-up before the disease fully develops.

Scaling this solution to industry, however, faces several challenges. Data availability is a major hurdle because the model depends on multiple years of consistent laboratory and questionnaire data per patient, which many health systems do not maintain in a structured, linked format, and combining records across hospitals raises interoperability and standardisation issues. Compute and engineering costs rise when ensemble methods (stacking, soft voting) are deployed at national scale across millions of patients, requiring robust MLOps pipelines for periodic retraining as population health patterns shift. Finally, ethical issues include patient data privacy when EHRs are pooled across institutions, the risk of the model performing unevenly across age groups, sexes, or ethnicities that were underrepresented in the original single-site training data, and the need for clear clinical governance so that a false negative forecast does not lead a patient to skip preventive screening they would otherwise have received.

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Answer:

(a) Three Key Insights Linked to CO4 Concepts

- Insight 1 - Inductive learning from longitudinal records: The article shows that inductive learning is not limited to a single snapshot of data; by pairing a patient's Year-Y attributes with their Year-(Y+1) outcome, the authors induce a general rule from many such labelled transitions and apply it to forecast a new patient's future state. This deepened my understanding of the Inductive Learning topic in CO4, because it demonstrates that the same generalise-from-labelled-examples principle taught for static classification can be extended to time-shifted, forecasting-style labels.
- Insight 2 - Statistical learning as an interpretable baseline against decision trees: Using Logistic Regression as the representative statistical model against which Random Forest, SVM, and XGBoost were benchmarked highlighted the practical trade-off covered under the Statistical Learning and Decision Trees topics in CO4: Logistic Regression is easier to interpret and audit clinically, but tree-based and ensemble models generally close the accuracy gap, echoing the pre-pruning versus post-pruning discussion of how model complexity trades off against interpretability.
- Insight 3 - Ensemble methods as a bridge to model robustness: Seeing that the ensemble techniques (confusion-matrix integration, stacking, soft voting) consistently outperformed any single model on cross-validation, but not necessarily on the single held-out test split, connected directly to the CO4 discussion of overfitting and generalisation: an ensemble reduces variance the way pruning reduces the variance of an individual decision tree, reinforcing why robustness should always be checked with cross-validation rather than a single train/test split.

(b) Proposed Extension

Novel experiment: I propose extending this work from a single-step forecasting classifier into a Reinforcement Learning-based sequential intervention recommender. Because the underlying dataset already contains multiple years of annual check-up records per patient, it is naturally suited to a Markov Decision Process formulation: each patient's yearly health snapshot becomes a state (Normal / Prediabetic / Diabetic, refined by FPG and HbA1c bands), the clinician's yearly recommendation (Lifestyle Counselling / Medication Review / Intensive Monitoring / No Action) becomes an action, and the reward could be defined as the improvement or delay in disease progression observed at the next annual visit. A Q-Learning or policy-gradient agent could then be trained on the historical multi-year trajectories to learn

which sequence of interventions minimises long-term progression to diabetes, rather than only predicting the next state passively.

Feasibility: This extension is practically feasible because the same six-year EHR structure already used for the classification study provides the sequential, multi-year state transitions that an RL agent needs, so no fundamentally new data collection would be required, only a redefinition of the labels into state-action-reward tuples and the addition of a simulated or logged intervention history.

Expected impact: If successful, this would shift the tool from a passive forecasting model into an active decision-support agent that recommends when and how strongly to intervene for a specific patient trajectory, which could reduce unnecessary medication for low-risk patients while intensifying monitoring for those on a fast-progressing trajectory, directly extending the paper's forward-prediction contribution into the Reinforcement Learning strand of CO4.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand